

Project Name :- Varley loop Test to Detect Underground cable fault.

Objectives :-

- i) To learn and verify the theoretical principles behind the varley loop test.
- ii) To determine the distance of a fault location from the test end in underground cable.

Theory :-

The Varley loop test is a technique used to locate faults (short circuits or earth faults) in underground cables. It is a reliable method for detecting of the fault location from the test end which is accessible for testing.

The test employs the wheatstone bridge principle to compare the resistance of two loops. One loop consists of a known good section

of the Cable under test, while the other loop includes a variable resistor. By adjusting variable resistor, the bridge is balanced, and the fault location can be calculated based on the resistance values. The necessary connections for the ground faults and short circuit fault are shown below:

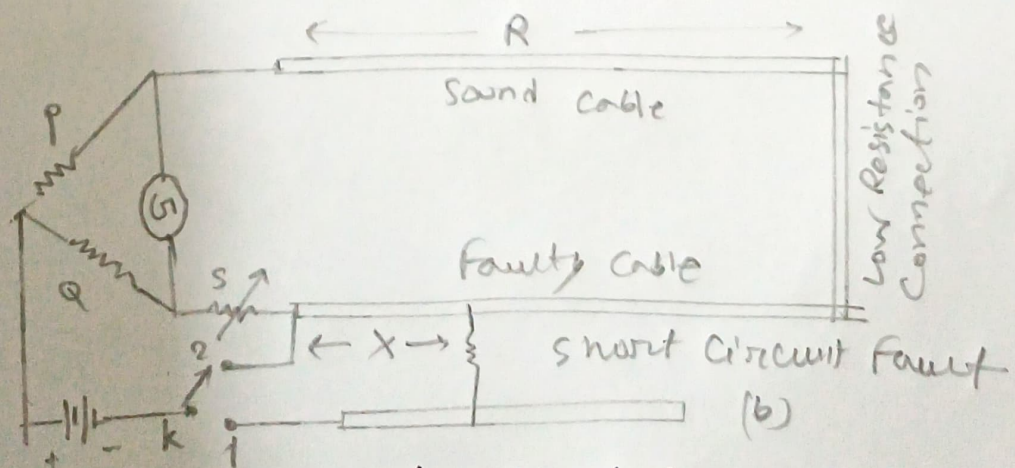
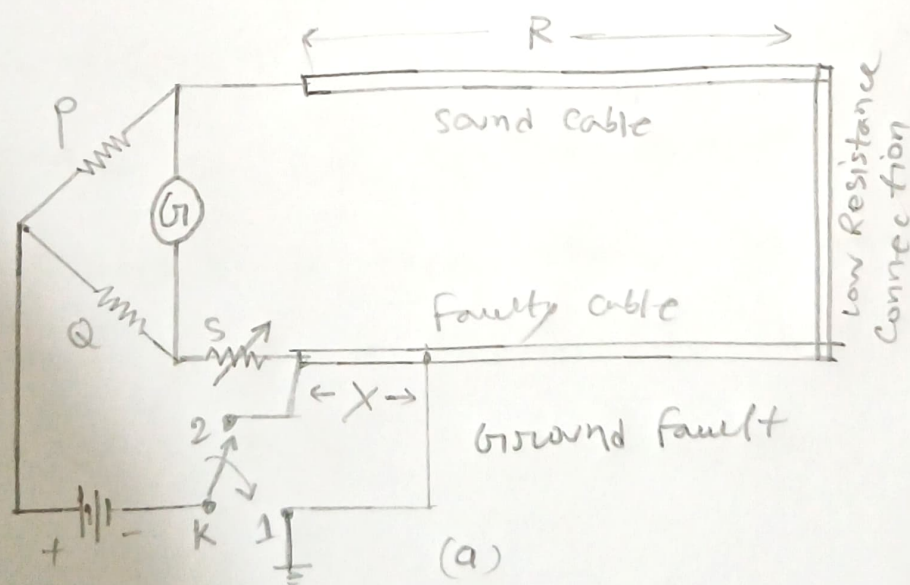


Fig (a, b) Varley loop test

A single pole double throw (SPDT) switch K is used in the circuit. switch K is first in position 2, and the resistance S is varied and balance obtained,

Let the value for S for balance be S_2 , the four arms of the wheatstone bridge are $P, Q, R+X, S_2$.

$$\therefore \text{At balance, } \frac{R+X}{S_2} = \frac{P}{Q} \dots \dots \dots (\text{eqn 5.1})$$

This determines $R+X$ as the total loop resistance as P, Q, S_2 are known.

The switch K is then thrown to position 1, and the bridge is rebalanced. Let the new value of S is S_1 . The four arms are now,

$P, Q, R, X+S_1$

$$\therefore \text{Now, } \frac{R}{X+S_1} = \frac{P}{Q}$$

$$\Rightarrow \frac{R+X+S_1}{X+S_1} = \frac{P+Q}{Q}$$

$$\Rightarrow (P+Q)(X+S_1) = (R+X+S_1)Q$$

$$\Rightarrow (S_1+X) = \frac{QR + QX + QS_1}{P+Q}$$

$$\Rightarrow X = \frac{RQ + XQ + QS_1}{P+Q} - S_1$$

$$\Rightarrow X = \frac{RQ + XQ + QS_1 - PS_1 - QS_1}{P+Q}$$

$$\Rightarrow X = \frac{Q(R+X) - PS_1}{P+Q}$$

$$\therefore X = \frac{Q(R+X) - PS_1}{P+Q} \dots \dots \dots (\text{eqn 5.2})$$

at balance, from eqn 5.1 $\Rightarrow$

$$\frac{P}{Q} = \frac{R+X}{S_2}$$

$$\Rightarrow (R+X)Q = PS_2 \dots \dots \dots (\text{eqn 5.3})$$

Putting eqn 5.3 in eqn 5.2 $\Rightarrow$

$$X = \frac{PS_2 - PS_1}{P+Q}$$

$$\boxed{X = \frac{P(S_2 - S_1)}{P+Q}} \dots \dots \dots (\text{eqn 5.4})$$

$\therefore$ Resistance from the test end to fault = X

Now, distance from the test end to fault

location, $\boxed{d = \frac{X}{r}} \dots \dots \dots \text{eqn 5.5}$

where, r is the per unit length of resistance

Apparatus :-

Table 5.1: Required apparatus for the project

SL No.	Name of the components	Specification	Quantity
01	Resistor	22 Ω , 10 Ω , 100 Ω tolerance = $\pm 5\%$	5, 1, 3
02	Rheostat	0 - 330 Ω	1
03	Female Banana socket	4 mm	12
04	Galvanometer	Analog, 110 V sensitivity: -300 to 300 μA	1
05	DC supply	5 V	1
06	Acrylic sheet	15 inch X 6 inch	1
07	Wood Box	18 inch X 8 inch X 3 inch	1
08	Connecting wire	male to male connectors	As required
09	Multimeter	DC: 200 mV to 600 V DC: 10 A R: 200 - 200 k Ω , 2 M Ω - 200 M Ω	1

Circuit Diagram :-

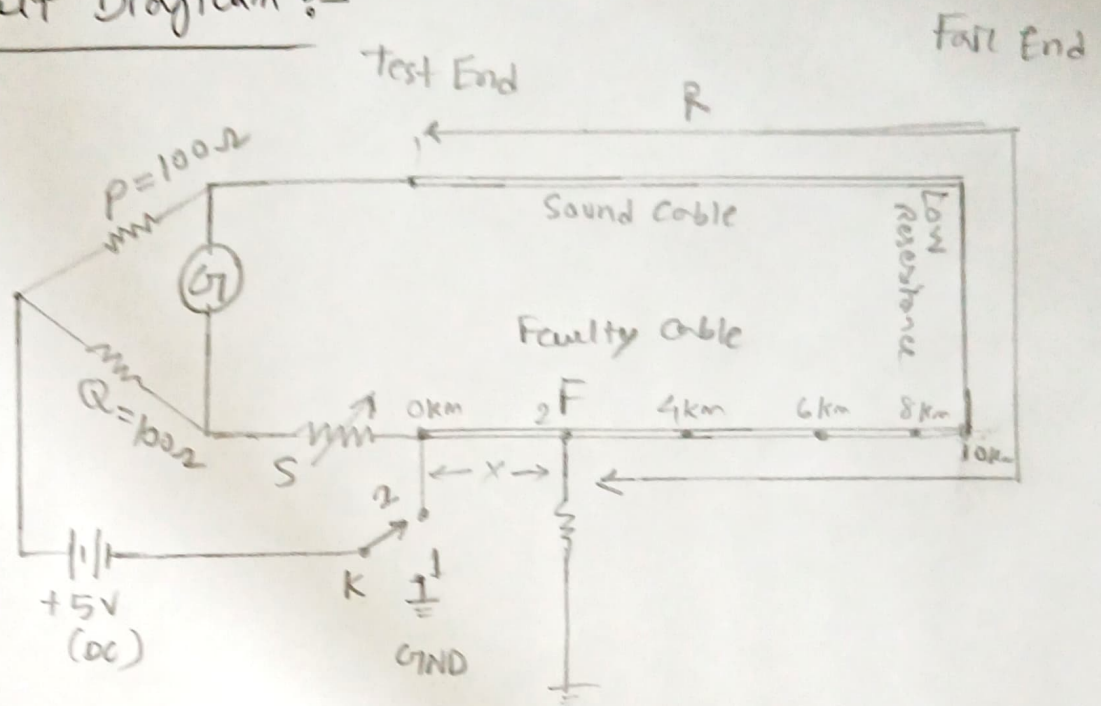


Fig 5.2 : Connection diagram of varley loop test

Data Table :-

Table 5.2: Data taken from the experiment

position	At balance (Resistance of S_2 when switch connected to 2)	Fault occurs when switch connected to S_1 (S_1 2)	Resistance from Test end to fault + X 2	Distance from the test end to fault $d = \frac{X}{2}$ km
At 2km	258.4	216.7	20.85	1.895
At 4km	258.4	173.5	42.45	3.859
At 6km	258.4	129.8	64.3	5.85
At 8km	258.4	85.5	86.45	7.859
At 10km	258.4	42.4	108	9.81

Calculation :-

When switch is connected to 2,

$$\frac{R+X}{S_2} = \frac{P}{Q}$$
$$\Rightarrow S_2 = \frac{P(R+X)}{P}$$

$$S_2 = 258.4 \Omega \quad [\text{Taken from the experiment}]$$

We know,

$$P=Q=100\Omega$$

$$r = 11\Omega/\text{km}$$

When fault occurs at 2 km,

$$S_1 = 216.7 \Omega$$

$$\therefore X_{2\text{km}} = \frac{P(S_2 - S_1)}{P+Q} = \frac{100(258.4 - 216.7)}{100+100}$$
$$= 20.85 \Omega$$

$$\therefore d = \frac{X}{r} = \frac{20.85}{11} = 1.895 \text{ km}$$

When fault occurs at 4 km,

$$S_1 = 173.5 \Omega$$

$$X_{4\text{km}} = \frac{P(S_2 - S_1)}{P+Q} = \frac{100(258.4 - 173.5)}{100+100}$$
$$= 42.45 \Omega$$

$$\therefore d = \frac{X}{r} = \frac{42.45}{11} = 3.859 \text{ km}$$

Similarly we can determine the fault location for 6 km, 8 km and 10 km.

Conclusion :-

The project successfully established to detect faults in underground cables. We established a circuit resembling a Varley loop test setup and created ground fault at specific location in the faulty cable. By adjusting the variable resistors and performing necessary calculations, we were able to determine the distance of the fault from the test end which is closely corresponded to the actual locations of the introduced faults, hence this validating the theoretical principles behind the test. In further exploration the test can be done with various faulty cables to assess the sensitivity in detecting fault of the kit. Also, there can be a comparison between other fault detection method (e.g. Murray loop test) to understand their relative advantages and sensitivity and accuracy. In conclusion, this project provided valuable insights into Varley loop test's functionality and potential impact on finding fault and repair process.